

Aligning Metabolomic Networks via Graph Learning for Common Subnetwork Discovery

Edwin Alvarez-Mamani¹, Deybis Paucar Pinto¹, Alfredo J. Ibañez¹, Madina Mansurova¹



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Problem formulation

The mass spectrometry analysis includes two approach:

- **Targeted approach:** the molecule of interest or m/z values (for MS and MS/MS) are predefined for quantitation of analytes present in a complex samples matrix.
- **Untargeted approach:** is the analysis to identify the unknowns in a sample, using parent ion and product or fragment ion analysis (MS/MS).

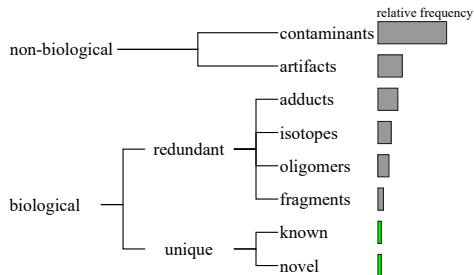


Figure 1: Composition of an untargeted metabolomic data.

Source: Adapted from [?]

Untargeted mass spectrometry workflow

A data matrix (sample vs. metabolites) acquired from a mass spectrometry-based metabolomics study includes not only biological information but also technical noise.

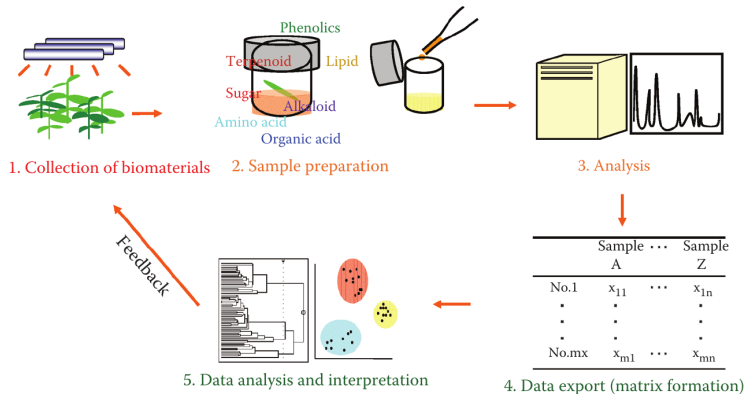


Figure 2: Metabolomics workflow.

Source: [?]

Our objectives

Aligning metabolomic networks for filter MS data.

Graph Neural Network (GNN)

It is a function $f: (A, X) \rightarrow Z$ that maps each node $v \in V$ of a graph $G = (V, E)$ to a low-dimensional embedding $z_v \in \mathbb{R}^d$ ($d \ll |V|$).

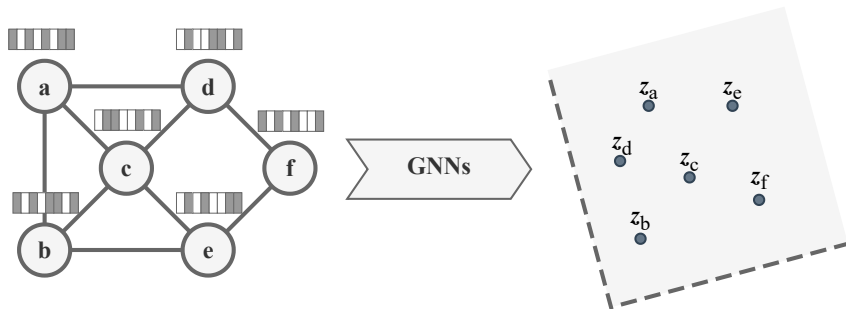


Figure 3: GNNs scheme.

Graph alignment problem

Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs. The goal is to align their nodes through a mapping function $\pi : \mathbf{V}_1 \rightarrow \mathbf{V}_2$, where each node $v \in V_1$ is mapped to its corresponding node $\pi(v) \in V_2$.

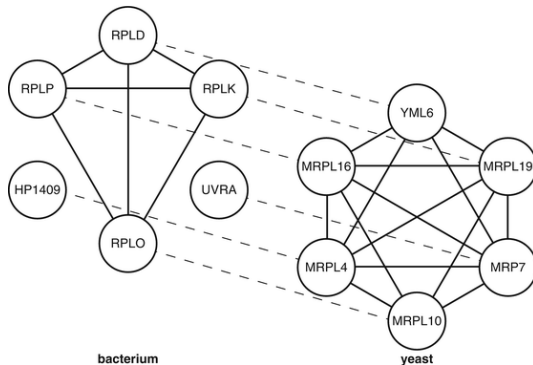


Figure 4: Node alignment.

Source: https://link.springer.com/rwe/10.1007/978-1-4419-9863-7_992

T-GAE: Transferable Graph Autoencoder for Network Alignment.
(He, J., Kanatsoulis, C., & Ribeiro, A., 2024)



Figure 5: T-GAE encoder.

Method

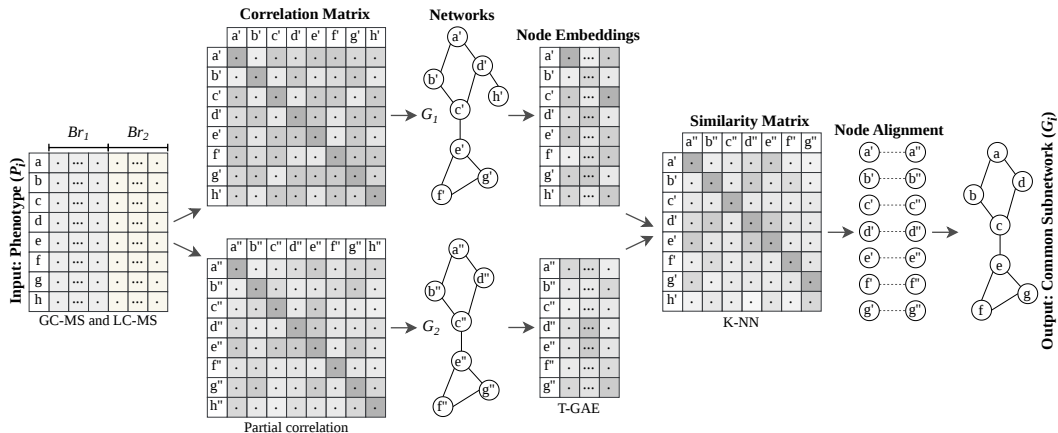


Figure 6: Our workflow.

- ① **Correlation estimation:** compute a partial correlation matrix.
- ② **Network construction:** build a metabolic network using a correlation threshold ($|\text{coef}| \geq 0.5$) and incorporate node attributes (R_t , M_z , and intensities).
- ③ **Graph representation learning:** learn node embeddings with the T-GAE architecture based on GIN and GINE layers.
- ④ **Network alignment:** identify metabolite correspondences using a KNN-based similarity matrix.

The experiments were run on an AMD Ryzen Threadripper PRO 5955WX computer with 128 cores, 256 GB of RAM, 2x NVIDIA RTX A6000 with 48 GB of VRAM. The source code was implemented in Python with PyTorch Geometric.

Experiments using the following datasets: Leaf dataset (citar), Cacao dataset.

Table 1: Cacao dataset.

Group	Subgroup	#Nodes	#Edges	Density	Is connected?	Node features (dim.)
SecoAmazonas	1	162	10268	0.787363	True	signal intensity (3)
SecoAmazonas	2	162	8728	0.669274	True	signal intensity (3)
SecoSanMartin	1	162	9771	0.749252	True	signal intensity (3)
SecoSanMartin	2	162	9705	0.744191	True	signal intensity (3)
SecoCusco	1	162	11297	0.866268	True	signal intensity (3)
SecoCusco	2	162	8765	0.672111	True	signal intensity (3)
FrescoAmazonas	1	162	9406	0.721264	True	signal intensity (3)
FrescoAmazonas	2	162	9006	0.690591	True	signal intensity (3)
FrescoSanMartin	1	162	9818	0.752856	True	signal intensity (3)
FrescoSanMartin	2	162	9454	0.724944	True	signal intensity (3)
FrescoCusco	1	162	10366	0.794878	True	signal intensity (3)
FrescoCusco	2	162	9249	0.709225	True	signal intensity (3)

Table 2: Experiments setup.

Features	Values
Node embedding	VGAE, LVGAE, ARGVA, DGI
Positional encoding	Laplacian eigenvector PE, Random walk PE
Dimensions	2, 3, 4, 8, 16, 32, 64, 128
Optimizer	Adam with $lr = 10^{-2}$, $weight_decay = 10^{-4}$
Early stopping	$patience = 10$
Epochs	100
Edge embedding	Weighted-L2
Outliers detection	COPOD with ($contamination = 0.1$)
Dataset	Synthetic, Mutant, Leaf, Mentos
Network variations	<i>none</i> (0:0), <i>one-to-one</i> (1:1), <i>many-to-many</i> (m:m)

Table 3: Dataset information.

Group	Subgroup	$ V $	$ E $	Density	Connected
<i>secoAmazonas</i>	1	2071	1 113 277	0.52	True
<i>secoAmazonas</i>	2	2071	1 085 089	0.51	True
<i>secoSanMartin</i>	1	2069	1 082 335	0.51	True
<i>secoSanMartin</i>	2	2071	1 103 292	0.51	True
<i>secoCusco</i>	1	2072	1 091 447	0.51	True
<i>secoCusco</i>	2	2072	1 086 182	0.51	True
<i>frescoCusco</i>	1	2071	1 114 859	0.52	True
<i>frescoCusco</i>	2	2072	1 089 225	0.51	True

Node embedding (T-GAE)

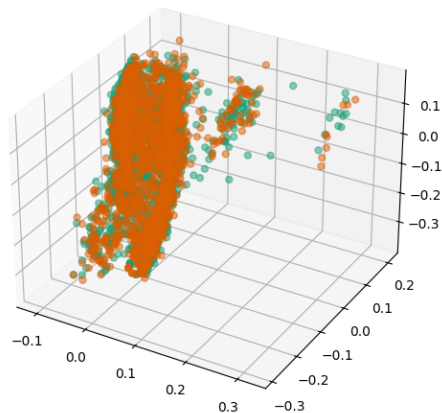


Figure 7: Node embedding (*secoSanMartin*).

Table 4: Node alignment (*secoSanMartin*).

ID 1	ID 2
13	13
19	252
27	378
31	1148
38	38
43	273
49	72
50	50
51	366
53	241
73	270
...	...
1825	1726

Table 5: Node filtered **133/2072** (*secoSanMartin*).

ID	Rt	Mz	Metabolite name
13	6.10	281.05	Unknown
38	6.16	182.10	Unknown
50	6.20	354.10	Unknown
128	6.47	119.20	Unknown
184	6.76	341.14	Unknown
261	7.23	371.06	Unknown
272	7.29	327.01	Unknown
314	7.61	57.02	DECANE; EI-B; MS
336	7.77	53.96	Unknown
339	7.81	328.01	Unknown
400	8.31	109.08	CITRONELLYL ACETATE...
402	8.31	109.13	BETA-HOMOCYCLOCITRAL...
407	8.37	401.11	Unknown

Our proposal, called GEMNA, is a new approach to the exploratory analysis of metabolic networks.

- We designed a pipeline for a metabolomic study. GEMNA consists of three steps: *network generation*, *network filtering*, and *similarity analysis*.
- We evaluated the execution time for generating node embeddings and edge embeddings. The experiments demonstrate that DGI model is faster than VGAE, LVGAE, ARGVA, to generate node embeddings.
- Experiments show that our GEMNA is an improvement over traditional statistical-based techniques. The evaluation by silhouette score reported that GEMNA obtained 0.41, meanwhile, the statistical method obtained -0.004 .
- We developed a web application for metabolomic network analysis and the source code is available on GitHub.

Based on these results, I recommend analyzing metabolomic networks using *Geometric Graph Neural Networks* (GGNN) models.

These models create node embeddings using the graph structure, node and edge features, and a feature tensor that helps infer the position of each metabolite within the network.

Thanks!
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edwin.alvarez@pucp.edu.pe
<https://github.com/win7>

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